Behavioral Sequences in App-Based Investment Frauds: An Experimental Behavioral-Economic Approach

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1 Introduction and Problem Statement

App-based investment frauds are often not a single deceptive event. Instead, fraudsters can gradually build trust, show fabricated investment returns, and introduce barriers that encourage victims to continue committing money [7, 1]. This suggests that investment fraud can be studied as a sequence of financial decisions rather than only as a victim/non-victim outcome.

Existing research has largely documented victim experiences, scam tactics, or susceptibility to hypothetical investment pitches, while fewer studies experimentally reproduce the sequential decision process within an interactive investment environment.

This project investigates: **How does sequential exposure to apparent rewards and withdrawal barriers alter subsequent financial decision-making, and do manipulative interfaces amplify this effect?** We model the process as a sequence in which a user first sees an apparent return, decides whether to reinvest, attempts to withdraw, encounters an artificial fee, and finally decides whether to make another payment.

The main goal is to identify points in this sequence where financial commitment increases and to examine whether backend reward manipulation and frontend interface manipulation affect escalation independently or interactively. Financial literacy and risk tolerance are also considered as individual differences that may affect susceptibility.

2 Behavioral Mechanisms and Hypotheses

The experiment is informed by behavioral economics and neuroeconomic theory, but it does not measure neural activity directly. Instead, it measures observable decisions that are consistent with theories of reward learning, loss aversion, sunk costs, and persuasion.

2.1 Reward Feedback

Fraudulent investment platforms may display fabricated or exaggerated returns to create an impression of investment success. Positive reward feedback can change the perceived attractiveness of continued investment. Reward-learning research provides a theoretical basis for studying how outcomes can influence subsequent decisions [8]. In this study, the effect is tested behaviorally by comparing participants receiving a 35% apparent return with those receiving a 0% return.

2.2 Withdrawal Barriers

Investment fraud may introduce artificial taxes, processing charges, or other payments when users attempt to withdraw their funds [7]. This creates a decision where abandoning an appar-

ently valuable balance may feel like accepting a loss. Such behavior is relevant to loss aversion and sunk-cost effects [6, 2]. The experiment therefore examines whether participants who have accumulated a larger apparent balance are more willing to pay an additional withdrawal fee.

2.3 Manipulative Interface

Online investment fraud can also use digital features to establish legitimacy, create urgency, and encourage continued engagement [1]. Scam communication can similarly use social and psychological tactics to build trust and commitment [3]. In this experiment, countdown timers, fabricated social proof, and visually salient progress/status cues are grouped into one **Manipulative UI** condition.

We do not directly measure physiological arousal, FOMO, regret, or heuristic processing. These are treated as possible interpretations of the observed behavior rather than direct measurements.

2.4 Hypotheses

- **H1 (Reward escalation):** Participants exposed to a 35% apparent return will show greater subsequent reinvestment than participants exposed to the 0% condition.
- **H2 (Withdrawal commitment):** Participants who have accumulated a larger apparent portfolio balance will be more likely to pay the withdrawal fee rather than abandon the funds.
- **H3 (Manipulative UI):** Participants exposed to the Manipulative UI will show greater reinvestment and a higher probability of paying the withdrawal fee than participants exposed to the Neutral UI.
- **H4 (Interaction):** The Manipulative UI will amplify the effect of positive reward feedback on subsequent escalation behavior, including reinvestment and willingness to pay the withdrawal fee.
- **H5 (Individual differences):** Financial literacy and risk tolerance will moderate susceptibility to the experimental manipulations [5, 4].

3 Methodology

We propose a 2×2 between-subjects factorial experiment:

$$\textbf{[Neutral UI vs. Manipulative UI]} \times \textbf{[0\% Gain vs. 35\% Gain]}.$$

Participants will be randomly assigned to one of the four conditions.

3.1 The 7-Stage Sequence

All participants will navigate the same simplified investment pipeline:

1. **Endowment:** The participant receives 1,000 in virtual starting capital.
2. **Initial Commitment:** The participant is asked to risk a baseline amount, such as 100.
3. **Reward Manipulation:** The system displays either a 0% return (control) or a 35% apparent profit (treatment).
4. **Reinvestment Measurement:** The participant chooses how much of the available balance to reinvest. The system records the decision latency and reinvestment amount.
5. **Withdrawal Attempt:** The participant attempts to withdraw the apparent balance.

6. **Withdrawal Barrier:** The system demands a 20% "tax/fee" based on the starting capital to unlock the funds.
7. **Final Decision:** The participant chooses whether to pay the fee or abandon the apparent funds.

3.2 Interface Conditions

The underlying financial sequence is identical across the two UI conditions.

- **Neutral UI:** A simple banking-style interface without urgency or social-validation cues.
- **Manipulative UI:** An interface containing countdown timers, fabricated social-proof messages, and visually salient progress/status cues.

The UI features are treated as a single experimental manipulation rather than as separate factors.

3.3 Secondary Measures

Along with the main behavioral outcomes, participants will provide short self-report ratings of:

- **Perceived trust:** how trustworthy and legitimate the platform appears.
- **Perceived risk:** how risky the participant considers the investment.
- **Decision latency:** time taken to make key financial decisions.

Trust and perceived risk are secondary measures intended to help interpret differences in behavior. They are not treated as direct measurements of underlying psychological or neural states.

Participants will also complete a brief validated financial-literacy measure and the relevant DOSPERT risk-taking items.

3.4 Incentives and Ethics

The virtual balances are linked to a real-world lottery for a 1,000 voucher, so that decisions have an economically meaningful consequence. The participant's final virtual balance determines their lottery entries, while paying the withdrawal fee reduces the virtual balance and therefore the number of entries. A fixed conversion rate between final virtual balance and lottery entries will be specified before data collection.

To reduce demand characteristics, participants will be told that the study concerns digital investment decision-making without being given the specific hypotheses beforehand. The simulated platform uses virtual funds, and participants will receive a full debriefing about the experimental manipulations immediately after participation.

The artificial withdrawal fee is a controlled analogue of withdrawal barriers used in investment fraud. The simplified simulation sacrifices some realism in exchange for experimental control and the ability to isolate specific behavioral mechanisms.

4 Quantitative Analysis

The primary outcomes are the post-reward reinvestment amount and the binary decision to pay the withdrawal fee.

- **Reinvestment:** A factorial OLS regression will estimate the effects of reward condition, UI condition, and their interaction on the reinvestment amount, while controlling for the initial commitment where appropriate.
- **Fee payment:** A logistic regression will model the probability of paying the Stage 6 withdrawal fee. The main test will be the **Reward** $\times$ **UI** interaction, which determines whether the Manipulative UI amplifies the effect of positive reward feedback.
- **Individual differences:** Financial literacy and DOSPERT risk-tolerance scores will be included as covariates. Interaction terms may be used to test whether these variables moderate the experimental effects.
- **Secondary measures:** Trust, perceived risk, and decision latency will be compared across conditions and explored in relation to reinvestment and fee-payment behavior.

The primary analysis will focus on the main effects of reward and UI condition and, most importantly, their interaction. Since the proposed protocol contains one main post-manipulation reinvestment decision, a mixed-effects model is not required unless the experiment is later expanded to collect repeated reinvestment decisions.

5 Expected Contribution

This study aims to provide a behavioral view of app-based investment fraud by examining the sequence of decisions that can lead from an apparent investment gain to additional financial commitment. By separating reward manipulation from interface manipulation, the experiment can test whether these mechanisms have independent or interacting effects on user behavior.

The results could also provide a basis for future fraud-prevention work, such as warnings at points of escalating commitment or safeguards around suspicious withdrawal fees.

References

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